

Tighter security for XMSS (Reduction-Based Approach)

Work in Progress

Abstract

Contents

1	Introduction	2
2	Chain Preimage Resistance	2
3	Adaptation of the security proof of Generalized XMSS	3
4	Analysis of SM-CHAIN in the ROM / QROM	4

DRAFT

1 Introduction

We attempt to reproduce the proof strategy of [1] (we reuse all its notations and constructions), with the following change in the security proof: Replace the final guessing argument (that uses SM-PRE and SM-UD, which leads to the $L \cdot v \cdot 2^w$ factor overhead) by a new (non-standard¹) property: Multi-Target Chain Preimage Resistance (SM-CHAIN).

It remains to prove a bound for SM-CHAIN in the ROM (doable) and QROM (not trivial).

2 Chain Preimage Resistance

Definition 2.1 (Multi-Target Chain Preimage Resistance). Let $\text{Th} : \mathcal{P} \times \mathcal{T} \times \mathcal{M} \rightarrow \mathcal{H}$ be a tweakable hash function with $\mathcal{H} \subseteq \mathcal{M}$. Let $w \geq 1$ be the log-chain-length, let $p \in \mathbb{N}$ be a query bound, and let $\mathcal{A}$ be a stateful algorithm. Consider the experiment $\text{SM-CHAIN}_{\text{Th},w,p}(\mathcal{A})$:

Setup:

1. Sample $P \xleftarrow{\$} \mathcal{P}$.
2. Initialize $Q := \emptyset$, $\text{Chains} := []$, and counter $\text{ctr} := 0$.

Query Phase: Run $\mathcal{A}$ with (classical) access to the Chain Oracle $\mathcal{O}_{\text{chain}}(T_1, \dots, T_{2^w-1})$ for $(T_1, \dots, T_{2^w-1}) \in \mathcal{T}^{2^w-1}$:

- (a) If $\text{ctr} \geq p$ or $|\{T_1, \dots, T_{2^w-1}\}| < 2^w - 1$ or $\{T_1, \dots, T_{2^w-1}\} \cap Q \neq \emptyset$, return $\perp$.
- (b) Sample $Y_0 \xleftarrow{\$} \mathcal{H}$.
- (c) Compute $Y_j := \text{Th}(P, T_j, Y_{j-1})$ for $j = 1, \dots, 2^w - 1$.
- (d) Append $((T_1, \dots, T_{2^w-1}), (Y_0, \dots, Y_{2^w-1}))$ to Chains .
- (e) Set $Q := Q \cup \{T_1, \dots, T_{2^w-1}\}$ and $\text{ctr} := \text{ctr} + 1$.
- (f) Return Y_{2^w-1} .

Reveal Phase: When $\mathcal{A}$ signals to proceed:

1. Disable $\mathcal{O}_{\text{chain}}$ and provide P to $\mathcal{A}$.
2. For each $j \in [|\text{Chains}|]$, initialize $\text{Revealed}_j := \emptyset$.
3. Give $\mathcal{A}$ (classical) access to the Reveal Oracle $\mathcal{O}_{\text{reveal}}(j, k)$ for $j \in [|\text{Chains}|]$, $k \in \{0, \dots, 2^w - 1\}$:
 - (a) Let $(Y_0^j, \dots, Y_{2^w-1}^j)$ be the chain values for chain j .
 - (b) Set $\text{Revealed}_j := \text{Revealed}_j \cup \{k\}$.
 - (c) Return Y_k^j .

Output: $\mathcal{A}$ outputs (j^*, x) with $j^* \in [|\text{Chains}|]$ and $x \in \mathcal{H}$.

Winning Condition: Let $((T_1^{j^*}, \dots, T_{2^w-1}^{j^*}), (Y_0^{j^*}, \dots, Y_{2^w-1}^{j^*}))$ be the j^* th entry in Chains . Let $k^* := \min(\text{Revealed}_{j^*} \cup \{2^w - 1\})$ be the earliest revealed position (or $2^w - 1$ if nothing was revealed). Output 1 if and only if $k^* \neq 0$ and:

$$\text{Th}(P, T_{k^*}^{j^*}, x) = Y_{k^*}^{j^*}.$$

¹The resulting security proof is thus not in the Standard Model (but ROM/QROM)

The advantage is:

$$\text{Adv}_{\text{Th},w,p}^{\text{SM-CHAIN}}(\mathcal{A}) := \Pr[\text{SM-CHAIN}_{\text{Th},w,p}(\mathcal{A}) \Rightarrow 1].$$

Intuition 2.2. The challenger constructs secret hash chains of the form:

$$Y_0 \xrightarrow{T_1} Y_1 \xrightarrow{T_2} Y_2 \xrightarrow{T_3} \dots \xrightarrow{T_{2^w-1}} Y_{2^w-1}$$

but initially only reveals the *final* value Y_{2^w-1} . Later, the attacker may query to see some intermediate values. The attacker wins by finding a preimage of the earliest revealed value in the chain.

3 Adaptation of the security proof of Generalized XMSS

Theorem 3.1 (Security of Generalized XMSS). *Consider the scheme $\text{SIG}[\text{IncEnc}, \text{Th}, K]$ with parameters $n, v, w, L, K \in \mathbb{N}$ and Th, IncEnc as in Construction 3. Then, for every algorithm $\mathcal{A}$, that makes no more than q_s signature queries, there are algorithms $\mathcal{B}_i$ with $T(\mathcal{A}) \approx T(\mathcal{B}_i)$ for all i and*

$$\begin{aligned} & \text{Adv}_{\text{SIG}[\text{IncEnc}, \text{Th}]}^{\text{SY-UF-CMA}}(\mathcal{A}) \\ & \leq \text{Adv}_{\text{Th}, 2 \cdot L \cdot v \cdot 2^w}^{\text{SM-TCR}}(\mathcal{B}_1) + \text{Adv}_{\text{IncEnc}, q_s}^{\text{T-COLL-RES}, K}(\mathcal{B}_2) + \text{Adv}_{\text{Th}, L \cdot v \cdot 2^w}^{\text{SM-TCR}}(\mathcal{B}_3) + 2 \cdot \text{Adv}_{\text{Th}, w, L \cdot v}^{\text{SM-CHAIN}}(\mathcal{B}_4). \end{aligned}$$

Proof. Up to Game.4, the proof is identical to [1] and is omitted here for brevity.

...

From now on, we can hence assume that

$$\text{Chain}_{\text{Th}, i^*, \text{ep}^*}(P, x_{i^*}^*, x_{i^*}^* - x_{i^*}^*, y_{i^*}^*) = y_{i^*}^*,$$

Game.5 (Final Reduction to Chain Preimage Resistance): We now complete the proof by reducing to multi-target chain preimage resistance. Recall that if Game.3 outputs 1, we have $x \neq x^*$ and there exists $j^* = \min\{j \in [v] : x_j^* < x_j\}$.

We construct a reduction $\mathcal{B}_4$ against $\text{SM-CHAIN}_{\text{Th}, w, L \cdot v}$ as follows:

1. **Query phase:** For each epoch $\text{ep} \in [L]$ and chain index $j \in [v]$, query $\mathcal{O}_{\text{chain}}(T_1^{j, \text{ep}}, \dots, T_{2^w-1}^{j, \text{ep}})$ to receive the final chain value $\text{pk}_{\text{ep}, j}$. Store the mapping from (ep, j) to the corresponding chain index in Chains . This uses $L \cdot v$ chain queries.
2. **After reveal phase:** Receive P from the game. Compute the Merkle tree from the one-time public keys $\text{pk}_{\text{ep}} = (\text{pk}_{\text{ep}, 1}, \dots, \text{pk}_{\text{ep}, v})$ for $\text{ep} \in [L]$ to obtain root . Set $\text{pk} := (\text{root}, P)$ and provide pk to $\mathcal{A}$.
3. **Simulating $\mathcal{O}_{\text{Sig}}(\text{ep}, m)$:**
 - (a) Compute the encoding $x = \text{IncEnc}(P, m, \rho, \text{ep})$ by sampling ρ appropriately (if after K attempts, no ρ is found, return $\perp$).
 - (b) For each $j \in [v]$: query $\mathcal{O}_{\text{reveal}}$ (chain index for $(\text{ep}, j), x_j$) to obtain y_j , the chain value at position x_j .
 - (c) Compute the Merkle path path_{ep} and return $\sigma := (\rho, (y_1, \dots, y_v), \text{path}_{\text{ep}})$.
4. **Extracting the forgery:** When $\mathcal{A}$ outputs $(\text{ep}^*, m^*, \sigma^*)$ with $\sigma^* = (\rho^*, (y_1^*, \dots, y_v^*), \text{path}_{\text{ep}^*}^*)$:
 - (a) Compute $x^* = \text{IncEnc}(P, m^*, \rho^*, \text{ep}^*)$ and find $j^* = \min\{j : x_j^* < x_j\}$.
 - (b) Compute $z' := \text{Chain}_{\text{Th}, j^*, \text{ep}^*}(P, x_{j^*}^*, x_{j^*}^* - x_{j^*}^* - 1, y_{j^*}^*)$.
 - (c) Output (chain index for $(\text{ep}^*, j^*), z')$.

The simulation is perfect, $T(\mathcal{B}_4) \approx T(\mathcal{A})$, and $\mathcal{B}_4$ wins the SM-CHAIN game whenever Game.3 outputs 1. Therefore:

$$\text{Adv}_{\text{SIG}}^{\text{Game.3}}(\mathcal{A}) \leq \text{Adv}_{\text{Th},w,L,v}^{\text{SM-CHAIN}}(\mathcal{B}_4).$$

Conclusion: Summing over all game hops, we obtain the desired result. $\square$

4 Analysis of SM-CHAIN in the ROM / QROM

Consider the simpler game, denoted Game.1, in which the the public parameter P is send to adversary directly after being sampled by the challenger (and no longer at the end of the reveal phase). We denote the advantage of an adversary $\mathcal{B}$ in this game by $\text{Adv}_{w,p}^{\text{Game.1}}(\mathcal{B})$. By trivial reduction, we have:

Lemma 4.1. *For any classical (resp. quantum) adversary $\mathcal{A}$ against SM-CHAIN, there exists a classical (resp. quantum) adversary $\mathcal{B}$ against Game.1 with $T(\mathcal{A}) \approx T(\mathcal{B})$ such that:*

$$\text{Adv}_{\text{Th},w,p}^{\text{SM-CHAIN}}(\mathcal{A}) \leq \text{Adv}_{\text{SM-CHAIN},w,p}^{\text{Game.1}}(\mathcal{B}).$$

Consider now Game.2, similar to Game.1 but in which the adversary doesn't have access to the Reveal Oracle. Furthermore, the chain length κ is no longer necessarily a power of two, minus one (previously it was of the form $2^w - 1$). Finally, we introduce a Corruption Oracle that enables the adversary to reveal all the hidded values of a chain, which naturally removes it from the candidates in the Winning Condition. For clarity, we describe Game.2 in full detail below.

Definition 4.2 (Game.2). Let $\text{Th} : \mathcal{P} \times \mathcal{T} \times \mathcal{M} \rightarrow \mathcal{H}$ be a tweakable hash function with $\mathcal{H} \subseteq \mathcal{M}$. Let $\kappa \geq 1$ be the chain-length, let $p \in \mathbb{N}$ be a query bound, and let $\mathcal{A}$ be a stateful algorithm. Consider the experiment $\text{GAME-2}_{\text{Th},\kappa,p}(\mathcal{A})$:

Setup:

1. Sample $P \xleftarrow{\$} \mathcal{P}$.
2. Initialize $Q := \emptyset$, $\text{Chains} := []$, $\text{Corrupted} := \emptyset$, and counter $\text{ctr} := 0$.

Query Phase: Run $\mathcal{A}$ with P and (classical) access to:

Chain Oracle $\mathcal{O}_{\text{chain}}(T_1, \dots, T_\kappa)$ for $(T_1, \dots, T_\kappa) \in \mathcal{T}^\kappa$:

- (a) If $\text{ctr} \geq p$ or $|\{T_1, \dots, T_\kappa\}| < \kappa$ or $\{T_1, \dots, T_\kappa\} \cap Q \neq \emptyset$, return $\perp$.
- (b) Sample $Y_0 \xleftarrow{\$} \mathcal{H}$.
- (c) Compute $Y_j := \text{Th}(P, T_j, Y_{j-1})$ for $j = 1, \dots, \kappa$.
- (d) Append $((T_1, \dots, T_\kappa), (Y_0, \dots, Y_\kappa))$ to Chains .
- (e) Set $Q := Q \cup \{T_1, \dots, T_\kappa\}$ and $\text{ctr} := \text{ctr} + 1$.
- (f) Return Y_κ .

Corrupt Oracle $\mathcal{O}_{\text{corrupt}}(j)$ for $j \in [|\text{Chains}|]$:

- (a) Let $((T_1^j, \dots, T_\kappa^j), (Y_0^j, \dots, Y_\kappa^j))$ be the j th entry in Chains .
- (b) Set $\text{Corrupted} := \text{Corrupted} \cup \{j\}$.
- (c) Return $(Y_0^j, \dots, Y_\kappa^j)$.

Output: $\mathcal{A}$ outputs (j^*, x) with $j^* \in [|\text{Chains}|]$ and $x \in \mathcal{H}$.

Winning Condition: Let $((T_1^{j^*}, \dots, T_\kappa^{j^*}), (Y_0^{j^*}, \dots, Y_\kappa^{j^*}))$ be the j^* -th entry in **Chains**. Output 1 if and only if $j^* \notin \text{Corrupted}$ and:

$$\text{Th}(P, T_\kappa^{j^*}, x) = Y_\kappa^{j^*}.$$

The advantage is:

$$\text{Adv}_{\text{Th}, \kappa, p}^{\text{GAME-2}}(\mathcal{A}) := \Pr[\text{GAME-2}_{\text{Th}, \kappa, p}(\mathcal{A}) \Rightarrow 1].$$

Lemma 4.3. *For any classical (resp. quantum) adversary $\mathcal{A}$ against $\text{GAME-1}_{\text{Th}, w, p}$, there exists $\kappa \in [1, 2^w - 1]$, and a classical (resp. quantum) adversary $\mathcal{B}$ against $\text{GAME-2}_{\text{Th}, \kappa, p}$ with $T(\mathcal{A}) \approx T(\mathcal{B})$ such that:*

$$\text{Adv}_{\text{Th}, w, p}^{\text{GAME-1}}(\mathcal{A}) \leq 2^w \cdot \text{Adv}_{\text{Th}, \kappa, p}^{\text{GAME-2}}(\mathcal{B})$$

Proof. Let $\kappa \in [1, 2^w - 1]$ be the most likely value of k^* (see the Winning Condition of Definition 2.1) among all winning executions of $\mathcal{A}$ in $\text{GAME-1}_{\text{Th}, w, p}$ (taking the smallest in case of equiprobability). In a winning scenario for $\mathcal{A}$, the value of k^* coincides with κ with probability at least $1/(2^w - 1) \geq 1/2^w$.

We construct the adversary $\mathcal{B}$ against $\text{GAME-2}_{\text{Th}, \kappa, p}$ as follows:

Setup: $\mathcal{B}$ receives the public parameter P from the GAME-2 challenger and forwards it to $\mathcal{A}$.

Simulating the Chain Oracle: When $\mathcal{A}$ queries $\mathcal{O}_{\text{chain}}(T_1, \dots, T_{2^w-1})$:

- (a) $\mathcal{B}$ queries its own chain oracle with $(T_1, \dots, T_\kappa)$ to receive Y_κ .
- (b) $\mathcal{B}$ extends the chain by computing:

$$Y_j := \text{Th}(P, T_j, Y_{j-1}) \quad \text{for } j = \kappa + 1, \dots, 2^w - 1.$$

- (c) $\mathcal{B}$ stores the mapping from this chain index (in $\mathcal{A}$'s view) to the corresponding index in GAME-2 , along with $(T_1, \dots, T_{2^w-1})$ and $(Y_\kappa, \dots, Y_{2^w-1})$.
- (d) $\mathcal{B}$ returns Y_{2^w-1} to $\mathcal{A}$.

Simulating the Reveal Oracle: When $\mathcal{A}$ queries $\mathcal{O}_{\text{reveal}}(j, k)$:

- (a) If $k \geq \kappa$: $\mathcal{B}$ retrieves the stored $Y_k^{(j)}$ and returns it to $\mathcal{A}$.
- (b) If $k < \kappa$: $\mathcal{B}$ queries $\mathcal{O}_{\text{corrupt}}(j)$ to obtain all chain values $(Y_0^{(j)}, \dots, Y_\kappa^{(j)})$, stores them, and returns $Y_k^{(j)}$ to $\mathcal{A}$. (Subsequent reveal queries on chain j can use these stored values.)

Extracting the Forgery: When $\mathcal{A}$ outputs (j^*, x) :

- (a) Let $k^* := \min(\text{Revealed}_{j^*} \cup \{2^w - 1\})$ be the earliest revealed position for chain j^* .
- (b) If $k^* \neq \kappa$: $\mathcal{B}$ **aborts** (wrong guess for κ).
- (c) If $k^* = \kappa$: $\mathcal{B}$ outputs (j^*, x) to its GAME-2 challenger.

The simulation of the game provided by the reduction is perfect, and both adversaries have similar running times. Furthermore, when $\mathcal{A}$ wins $\text{GAME-1}_{\text{Th}, w, p}$, there is at least a $1/2^w$ probability that $\mathcal{B}$ does not abort and wins $\text{GAME-2}_{\text{Th}, \kappa, p}$. This concludes the proof. $\square$

We now define **Game.3**, which is the same as **Game.2** except that we remove the **Corrupt Oracle** and modify the **Winning Condition** accordingly.

Lemma 4.4. *If Th is modeled as a random oracle, for any classical (resp. quantum) adversary $\mathcal{A}$ against $\text{GAME-2}_{\text{Th},\kappa,p}$, there exists a classical (resp. quantum) adversary $\mathcal{B}$ against $\text{GAME-3}_{\text{Th},\kappa,p}$ with $T(\mathcal{A}) \approx T(\mathcal{B})$ such that:*

$$\text{Adv}_{\text{Th},\kappa,p}^{\text{GAME-2}}(\mathcal{A}) \leq \text{Adv}_{\text{Th},\kappa,p}^{\text{GAME-3}}(\mathcal{B})$$

Proof. TODO This is very intuitive since with a RO, querying the Corrupt Oracle does not provide any information. But how to proceed formally? $\square$

For clarity, we give a full definition of Game.3 below.

Definition 4.5 (Game.3). Let $\text{Th} : \mathcal{P} \times \mathcal{T} \times \mathcal{M} \rightarrow \mathcal{H}$ be a tweakable hash function with $\mathcal{H} \subseteq \mathcal{M}$. Let $\kappa \geq 1$ be the chain-length, let $p \in \mathbb{N}$ be a query bound, and let $\mathcal{A}$ be a stateful algorithm. Consider the experiment $\text{GAME-3}_{\text{Th},\kappa,p}(\mathcal{A})$:

Setup:

1. Sample $P \xleftarrow{\$} \mathcal{P}$.
2. Initialize $Q := \emptyset$, $\text{Chains} := []$, and counter $\text{ctr} := 0$.

Query Phase: Run $\mathcal{A}$ with P and (classical) access to:

Chain Oracle $\mathcal{O}_{\text{chain}}(T_1, \dots, T_\kappa)$ for $(T_1, \dots, T_\kappa) \in \mathcal{T}^\kappa$:

- (a) If $\text{ctr} \geq p$ or $|\{T_1, \dots, T_\kappa\}| < \kappa$ or $\{T_1, \dots, T_\kappa\} \cap Q \neq \emptyset$, return $\perp$.
- (b) Sample $Y_0 \xleftarrow{\$} \mathcal{H}$.
- (c) Compute $Y_j := \text{Th}(P, T_j, Y_{j-1})$ for $j = 1, \dots, \kappa$.
- (d) Append $((T_1, \dots, T_\kappa), (Y_0, \dots, Y_\kappa))$ to Chains .
- (e) Set $Q := Q \cup \{T_1, \dots, T_\kappa\}$ and $\text{ctr} := \text{ctr} + 1$.
- (f) Return Y_κ .

Output: $\mathcal{A}$ outputs (j^*, x) with $j^* \in [|\text{Chains}|]$ and $x \in \mathcal{H}$.

Winning Condition: Let $((T_1^{j^*}, \dots, T_\kappa^{j^*}), (Y_0^{j^*}, \dots, Y_\kappa^{j^*}))$ be the j^* th entry in Chains . Output 1 if and only if:

$$\text{Th}(P, T_\kappa^{j^*}, x) = Y_\kappa^{j^*}.$$

The advantage is:

$$\text{Adv}_{\text{Th},\kappa,p}^{\text{GAME-3}}(\mathcal{A}) := \Pr[\text{GAME-3}_{\text{Th},\kappa,p}(\mathcal{A}) \Rightarrow 1].$$

References

- [1] J. Drake, D. Khovratovich, M. Kudinov, and B. Wagner, “Hash-based multi-signatures for post-quantum ethereum,” Cryptology ePrint Archive, Paper 2025/055, 2025. [Online]. Available: <https://eprint.iacr.org/2025/055>